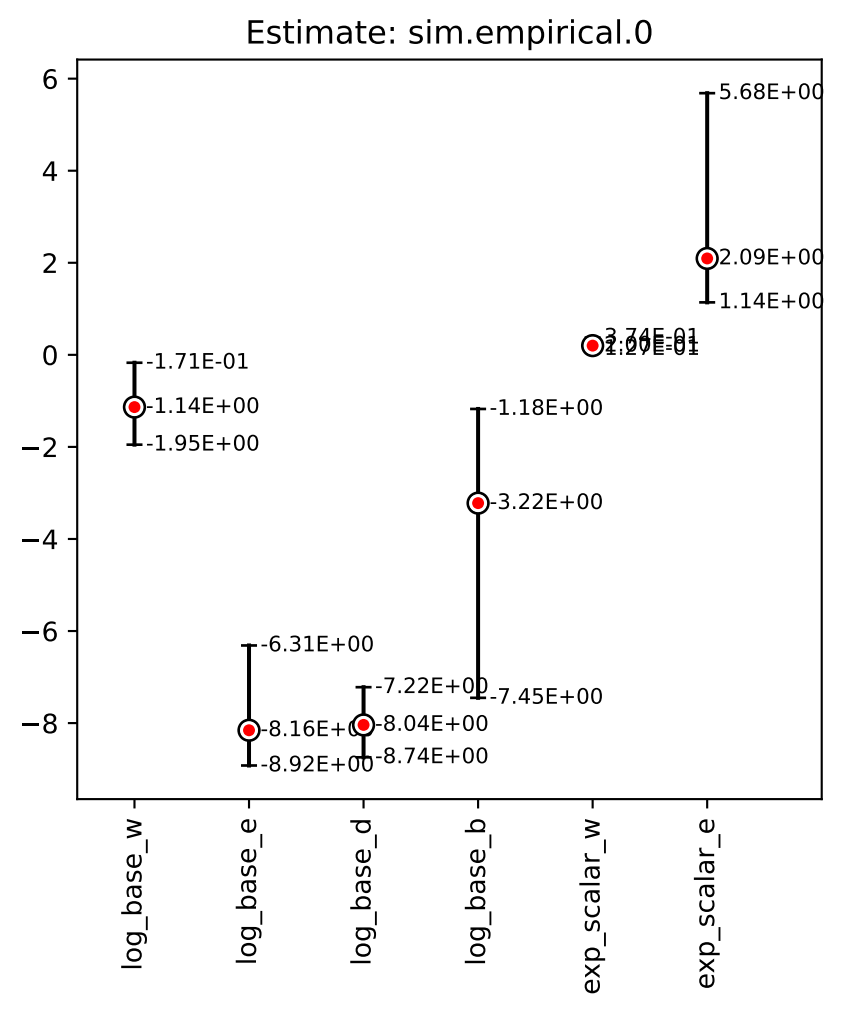
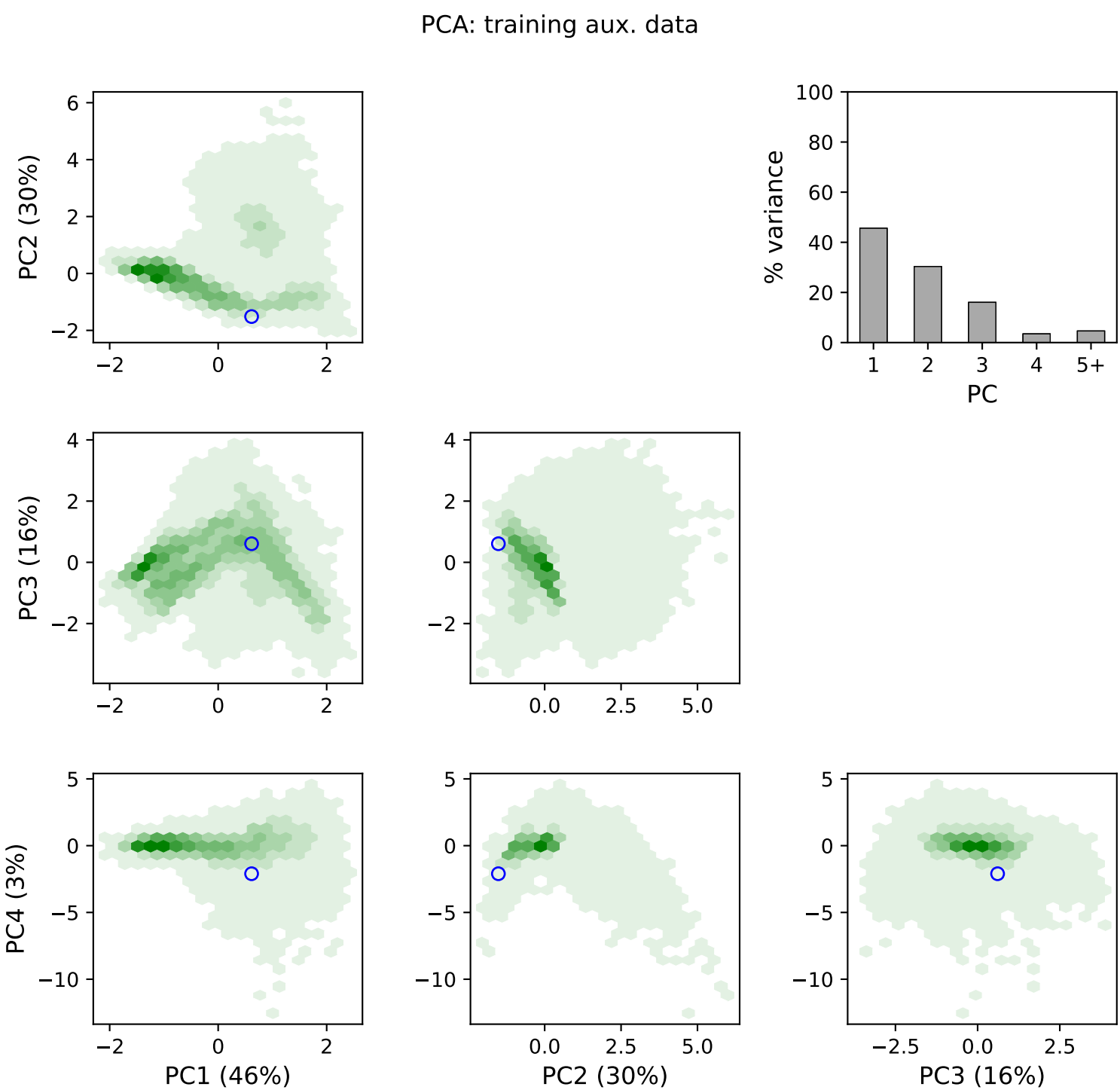
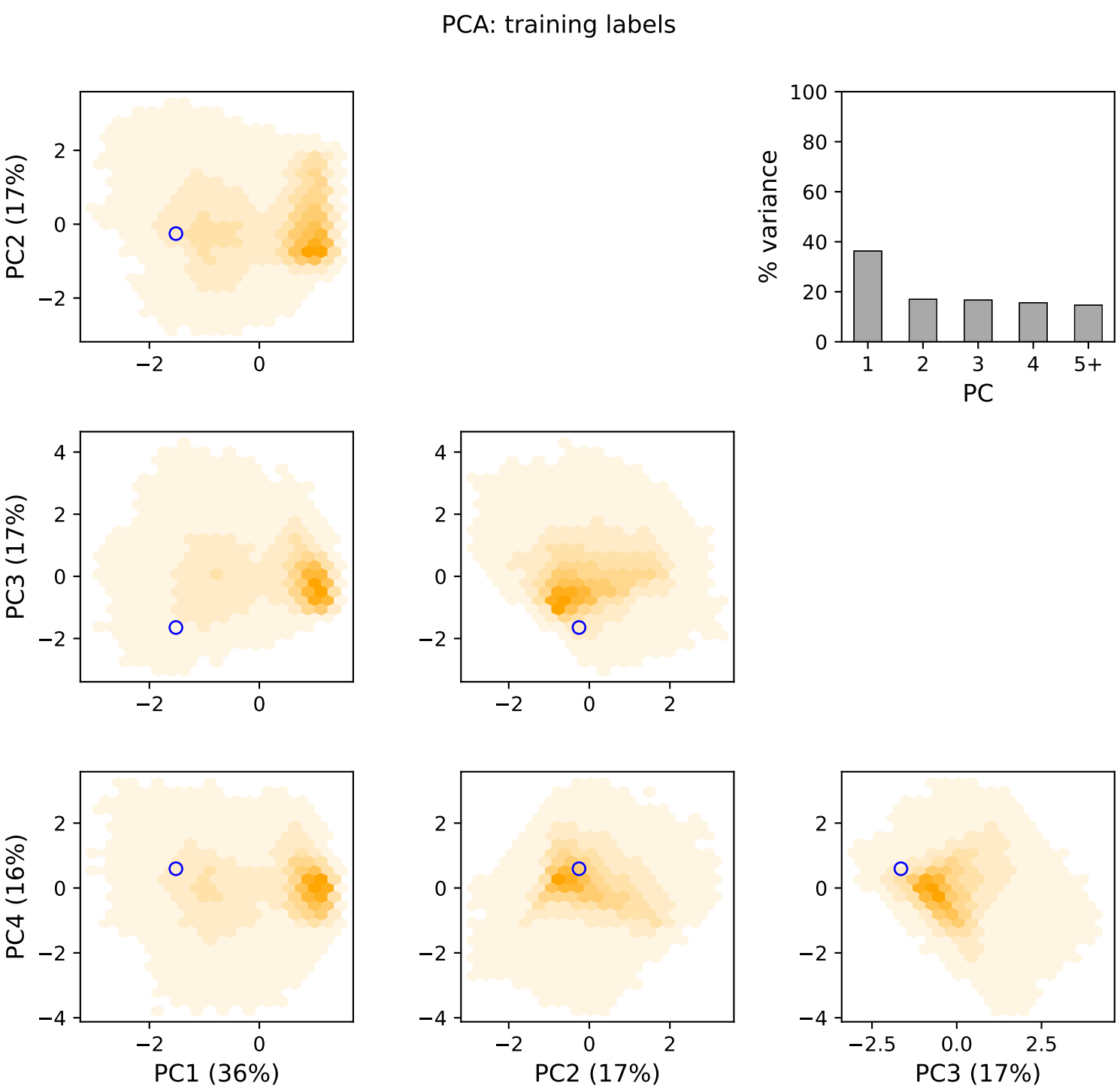
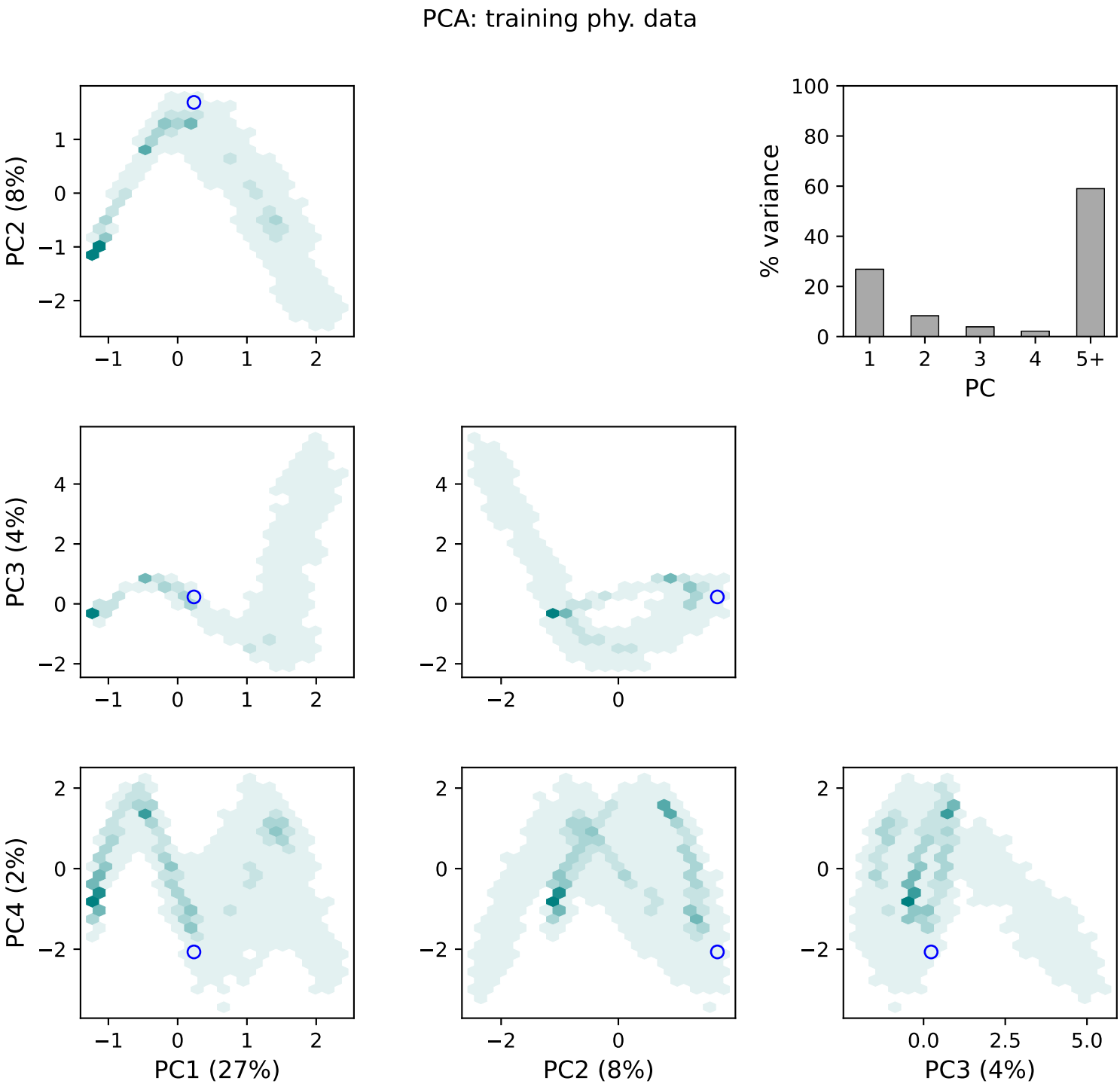
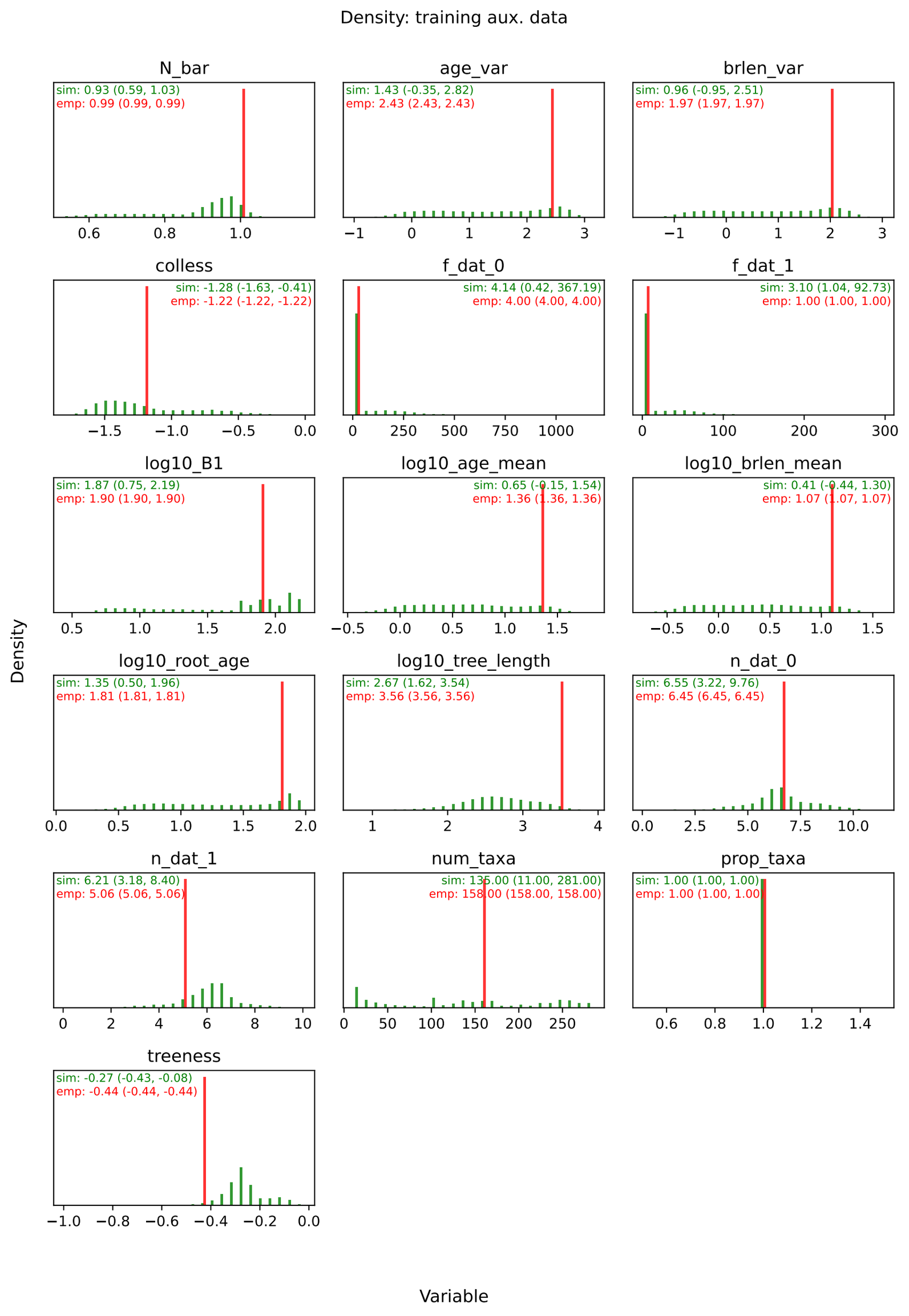
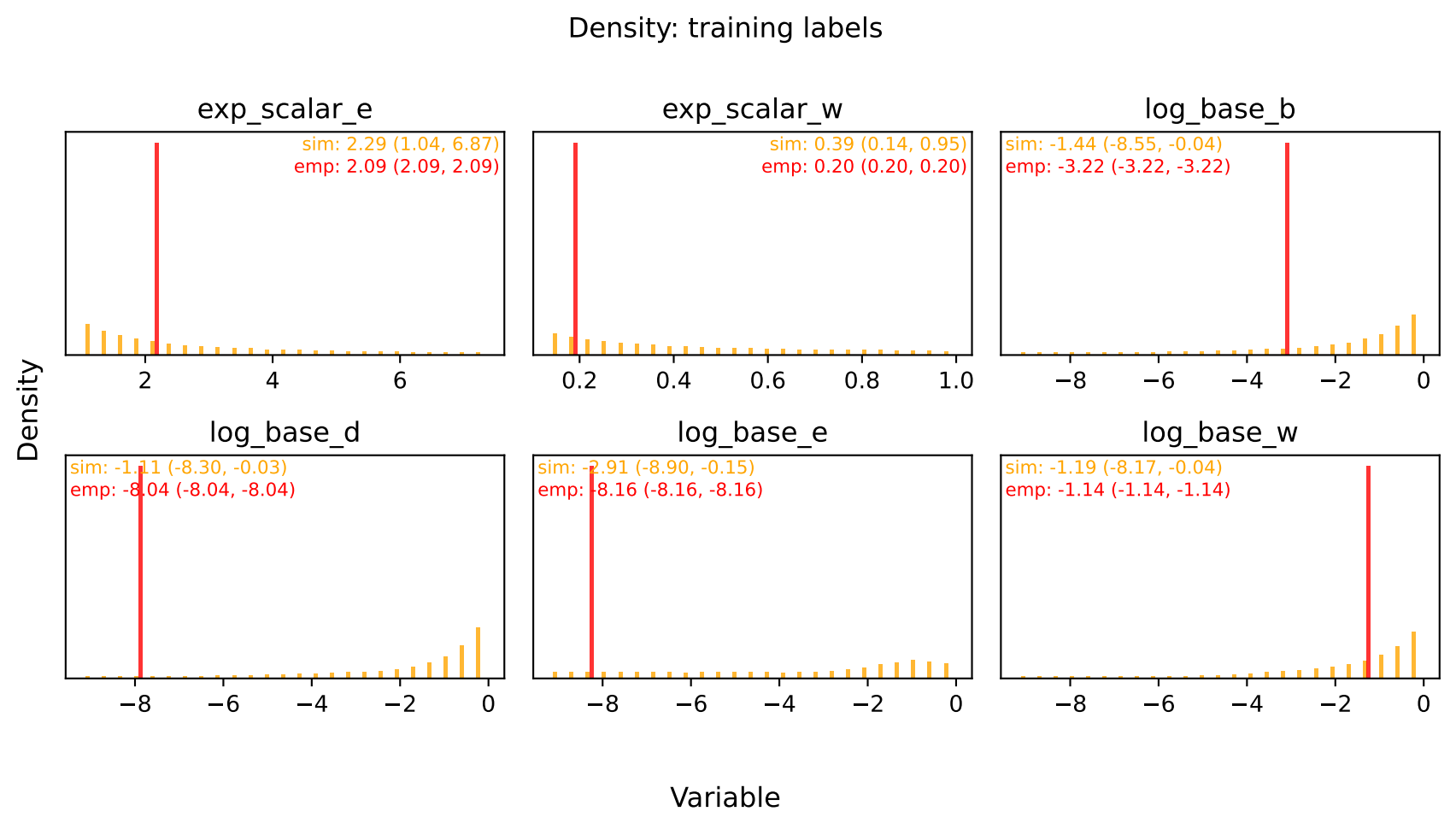

Estimate: sim.empirical.0
6
4
2
0
−2
−4
−6
−8
-1.71E-01
-1.14E+00
-1.95E+00
-6.31E+00
-8.16E+00
-8.92E+00
-7.22E+00
-8.04E+00
-8.74E+00
-1.18E+00
-3.22E+00
-7.45E+00
3.74E-01
5.68E+00
2.09E+00
1.14E+00
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w
exp_scalar_e

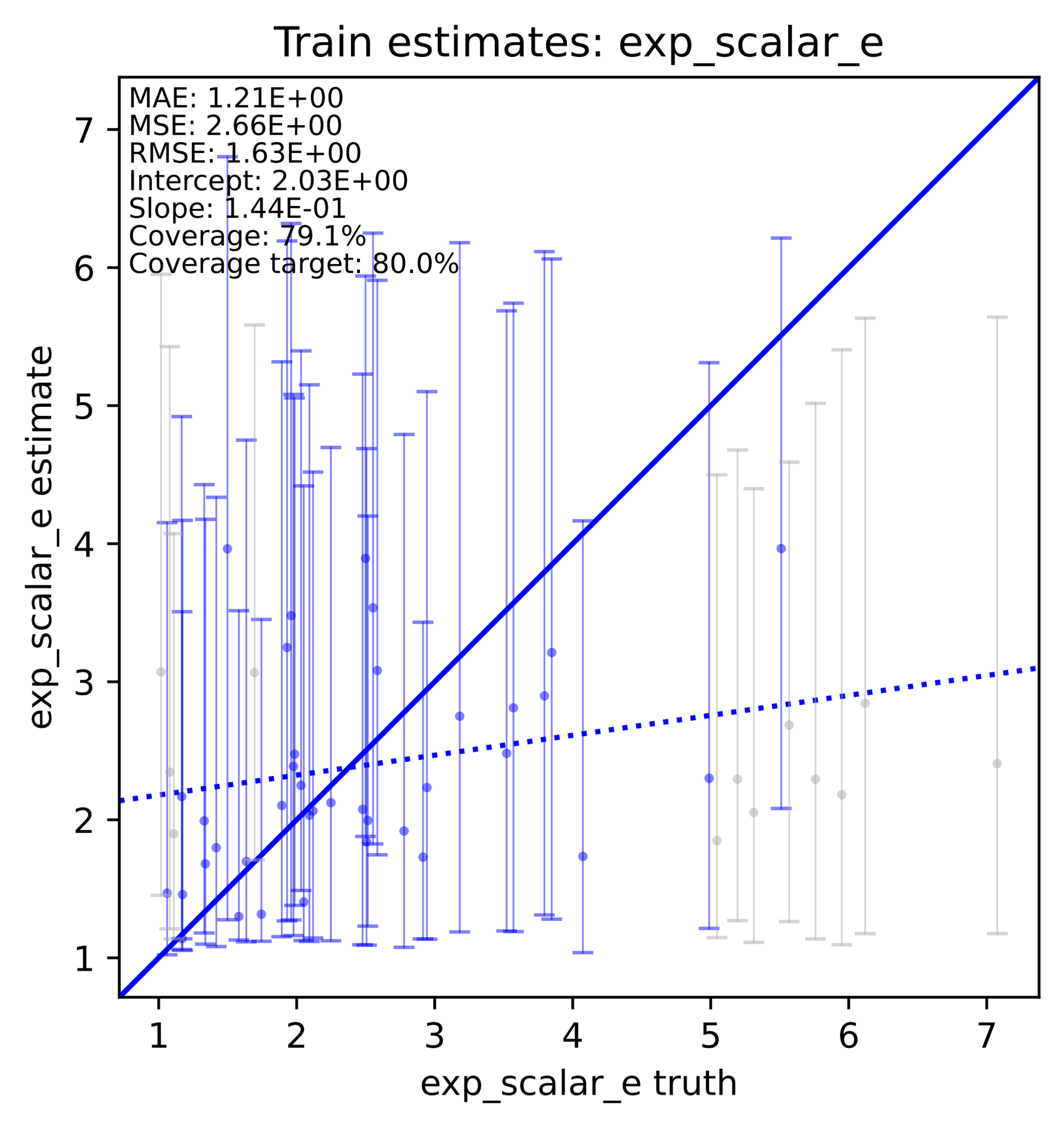

PCA: training aux. data
PC2 (30%)
% variance
PC
PC3 (16%)
PC4 (3%)
PC1 (46%)
PC2 (30%)
PC3 (16%)

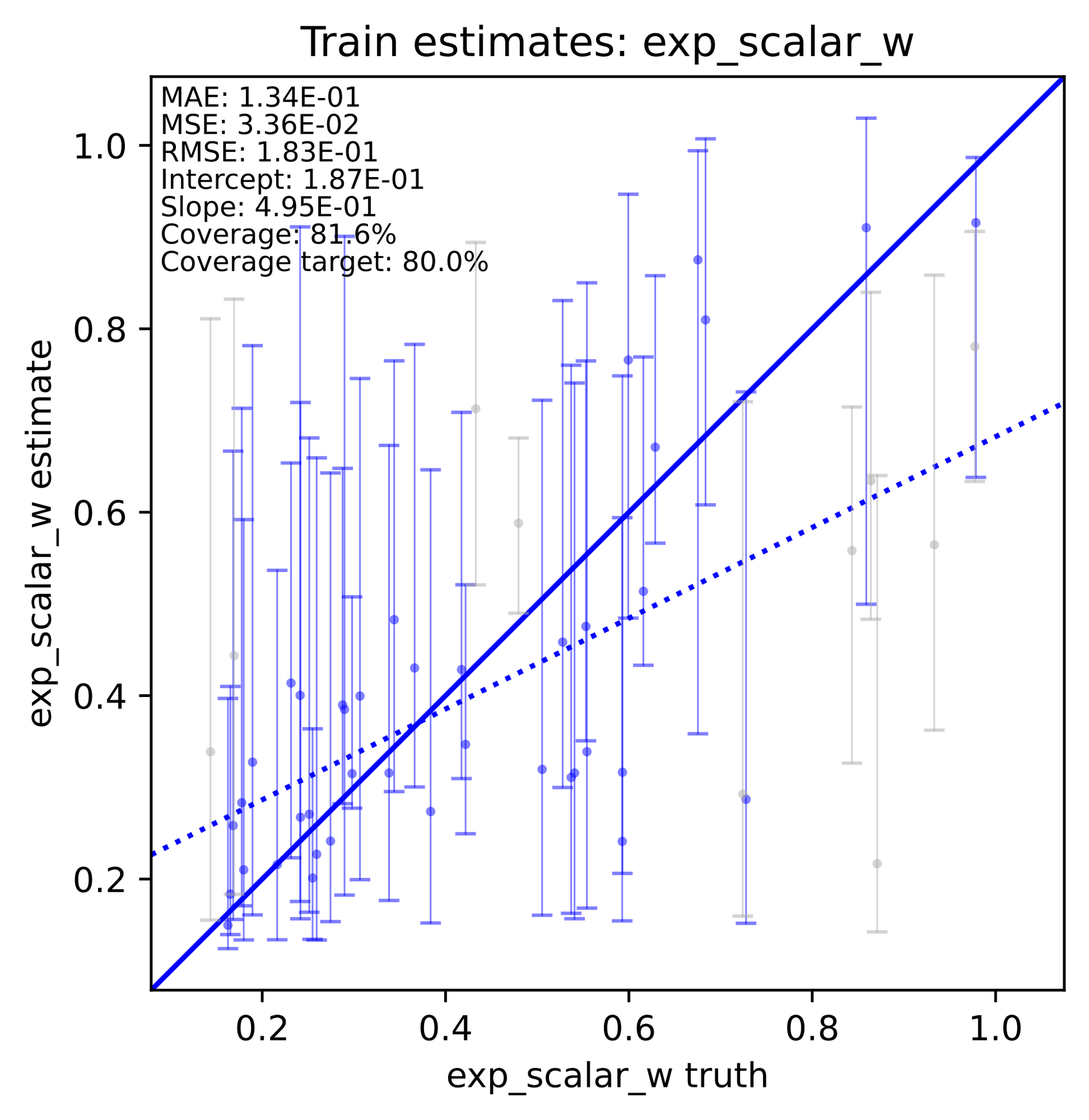

PCA: training labels
PC2 (17%)
% variance
PC
PC3 (17%)
PC4 (16%)
PC1 (36%)
PC2 (17%)
PC3 (17%)

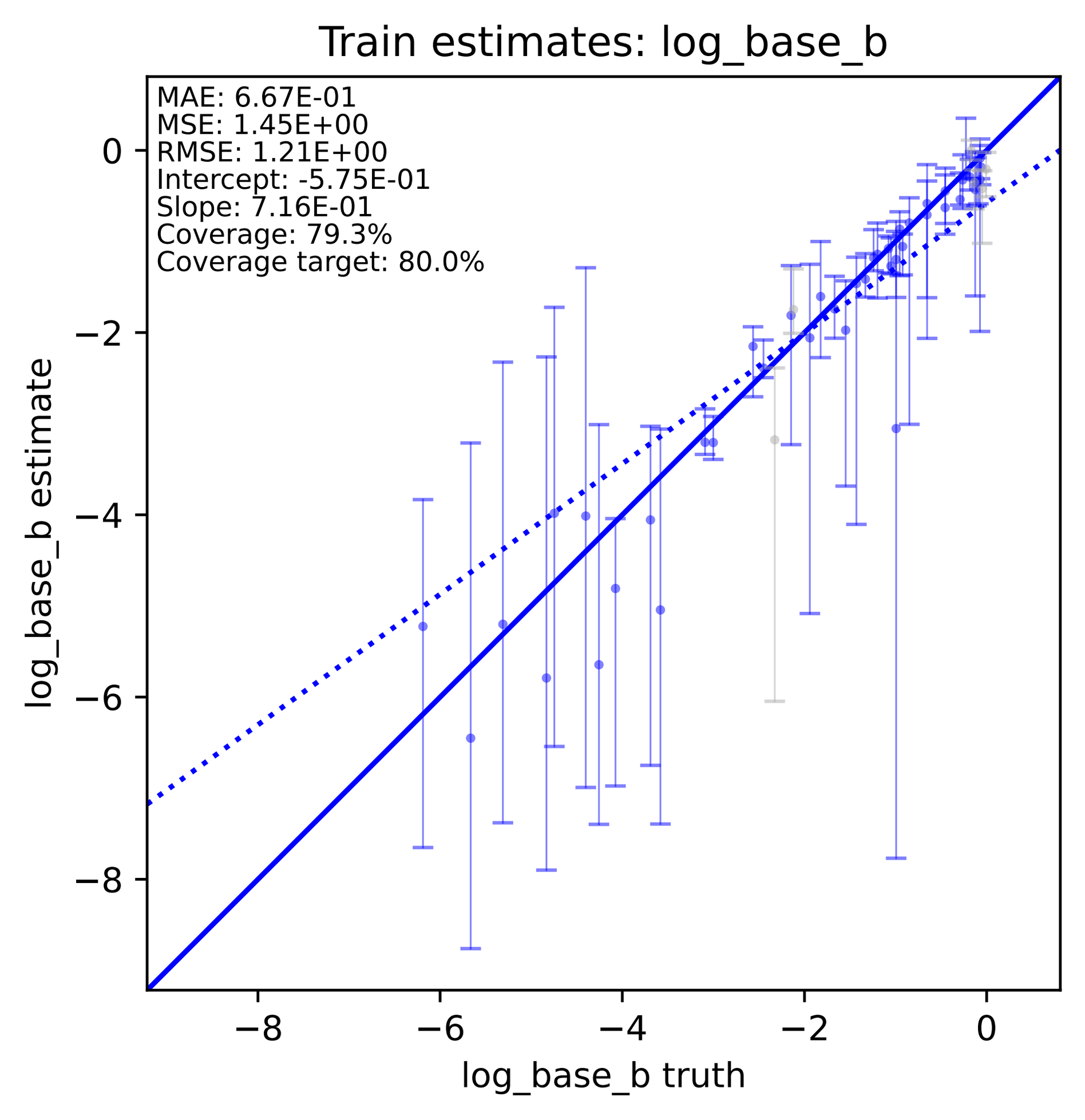

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (27%)
PC2 (8%)
PC3 (4%)

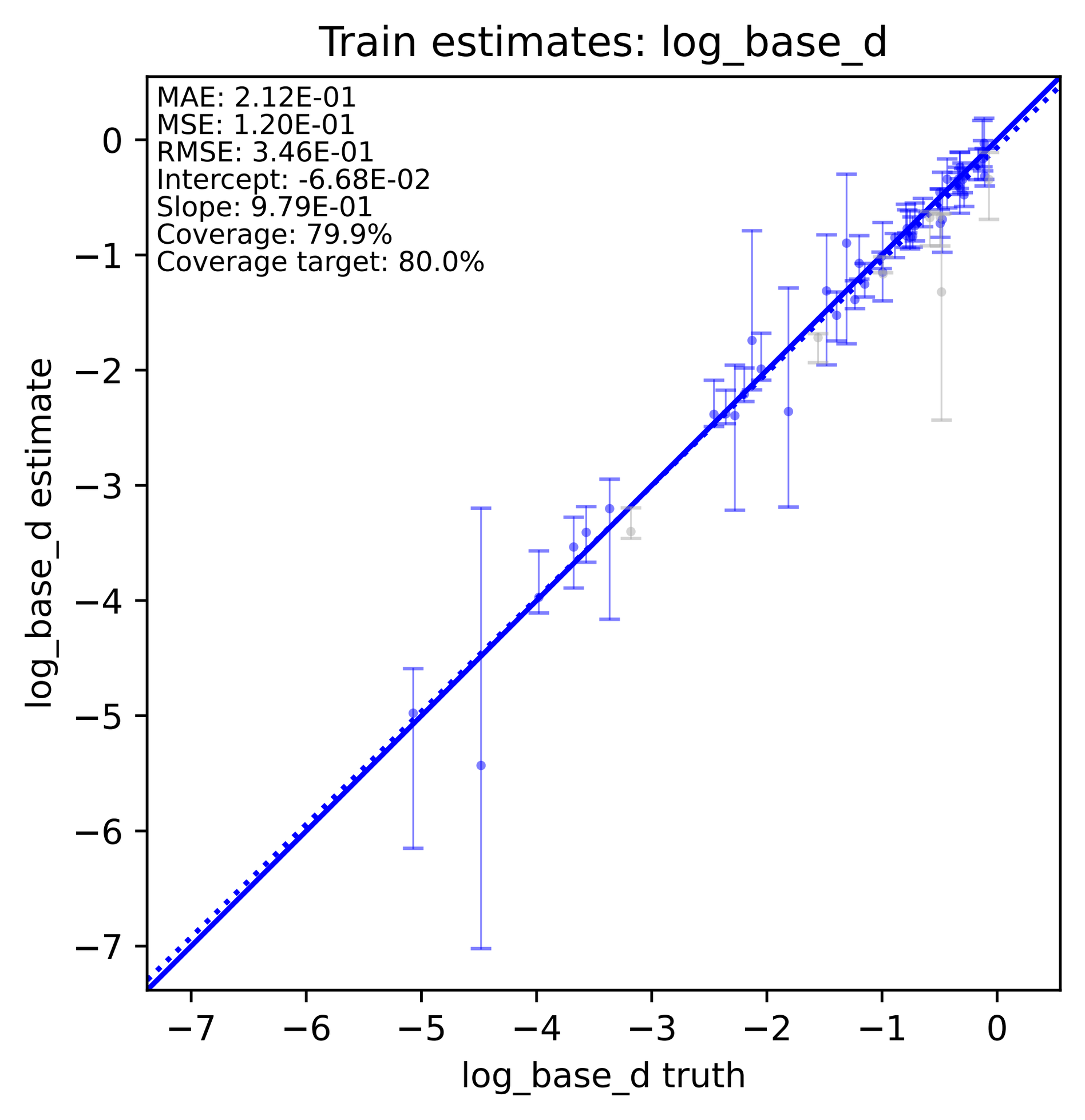

Density: training aux. data
N_bar
sim: 0.93 (0.59, 1.03)
emp: 0.99 (0.99, 0.99)
age_var
sim: 1.43 (-0.35, 2.82)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.96 (-0.95, 2.51)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.28 (-1.63, -0.41)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 4.14 (0.42, 367.19)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 3.10 (1.04, 92.73)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.87 (0.75, 2.19)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.65 (-0.15, 1.54)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.41 (-0.44, 1.30)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.35 (0.50, 1.96)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.67 (1.62, 3.54)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.55 (3.22, 9.76)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 6.21 (3.18, 8.40)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 135.00 (11.00, 281.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.27 (-0.43, -0.08)
emp: -0.44 (-0.44, -0.44)
Density
Variable

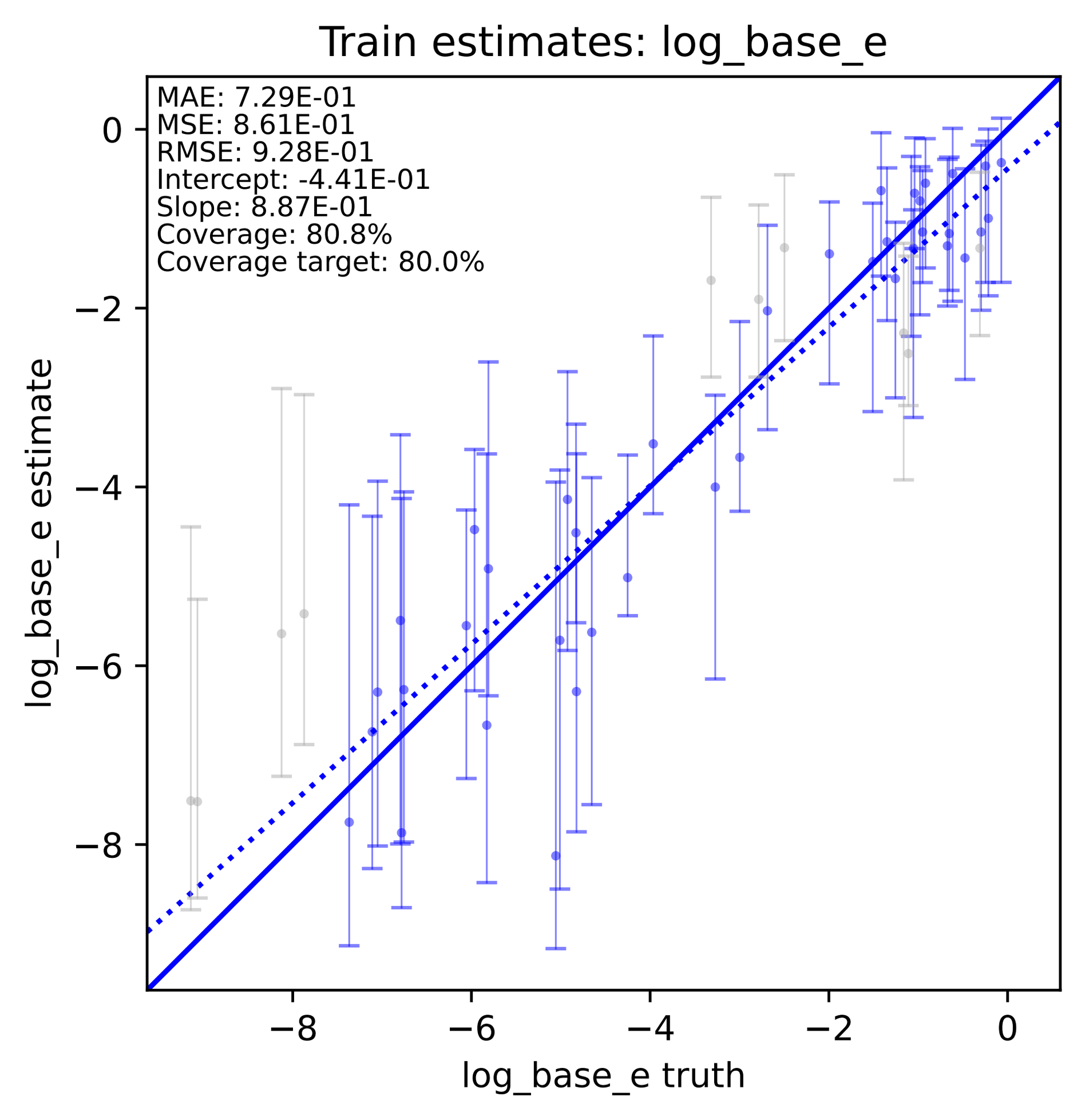

Density: training labels
exp_scalar_e
sim: 2.29 (1.04, 6.87)
emp: 2.09 (2.09, 2.09)
exp_scalar_w
sim: 0.39 (0.14, 0.95)
emp: 0.20 (0.20, 0.20)
log_base_b
sim: -1.44 (-8.55, -0.04)
emp: -3.22 (-3.22, -3.22)
log_base_d
sim: -1.11 (-8.30, -0.03)
emp: -8.04 (-8.04, -8.04)
log_base_e
sim: -2.91 (-8.90, -0.15)
emp: -8.16 (-8.16, -8.16)
log_base_w
sim: -1.19 (-8.17, -0.04)
emp: -1.14 (-1.14, -1.14)
Density
Variable

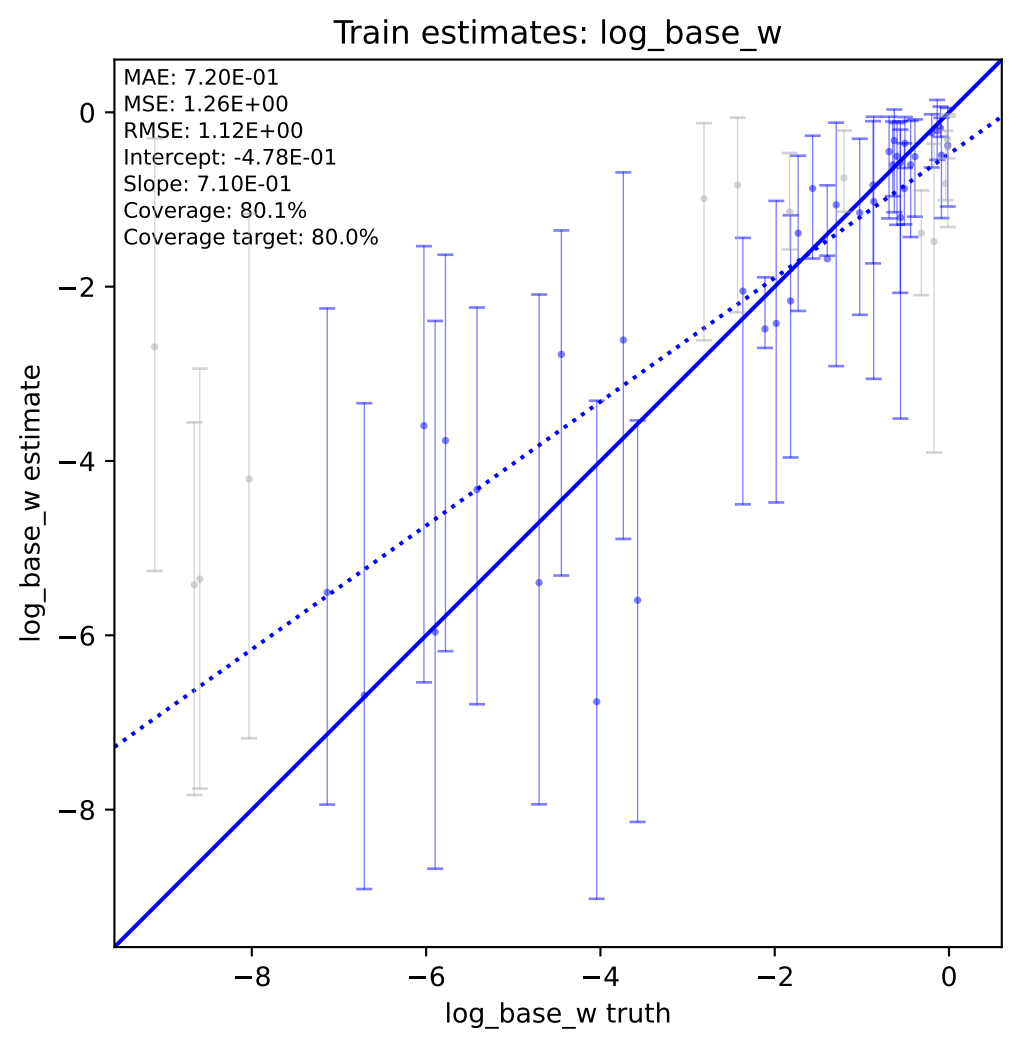

Train estimates: exp_scalar_e
MAE: 1.21E+00
MSE: 2.66E+00
RMSE: 1.63E+00
Intercept: 2.03E+00
Slope: 1.44E-01
Coverage: 79.1%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

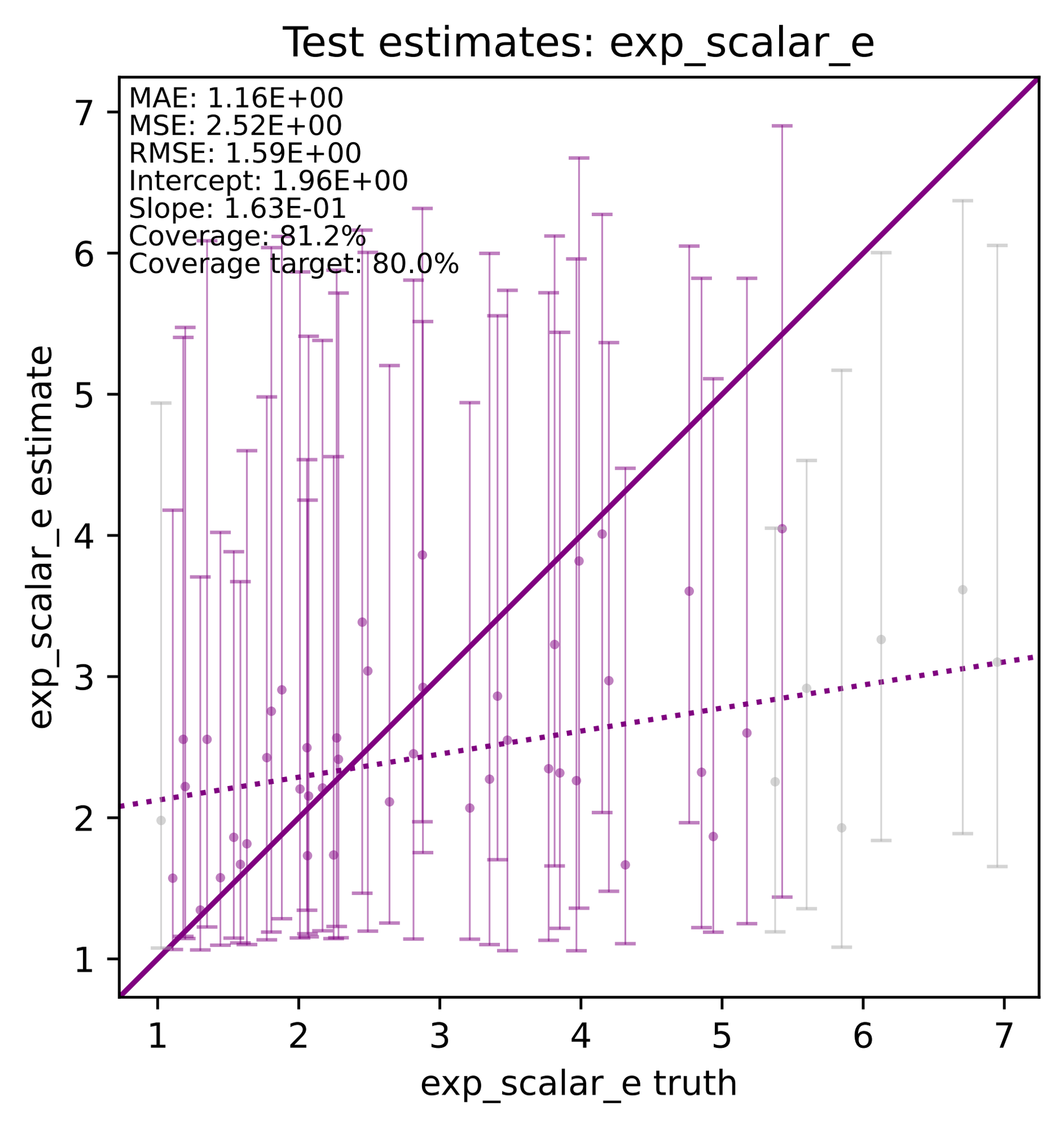

Train estimates: exp_scalar_w
MAE: 1.34E-01
MSE: 3.36E-02
RMSE: 1.83E-01
Intercept: 1.87E-01
Slope: 4.95E-01
Coverage: 81.6%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

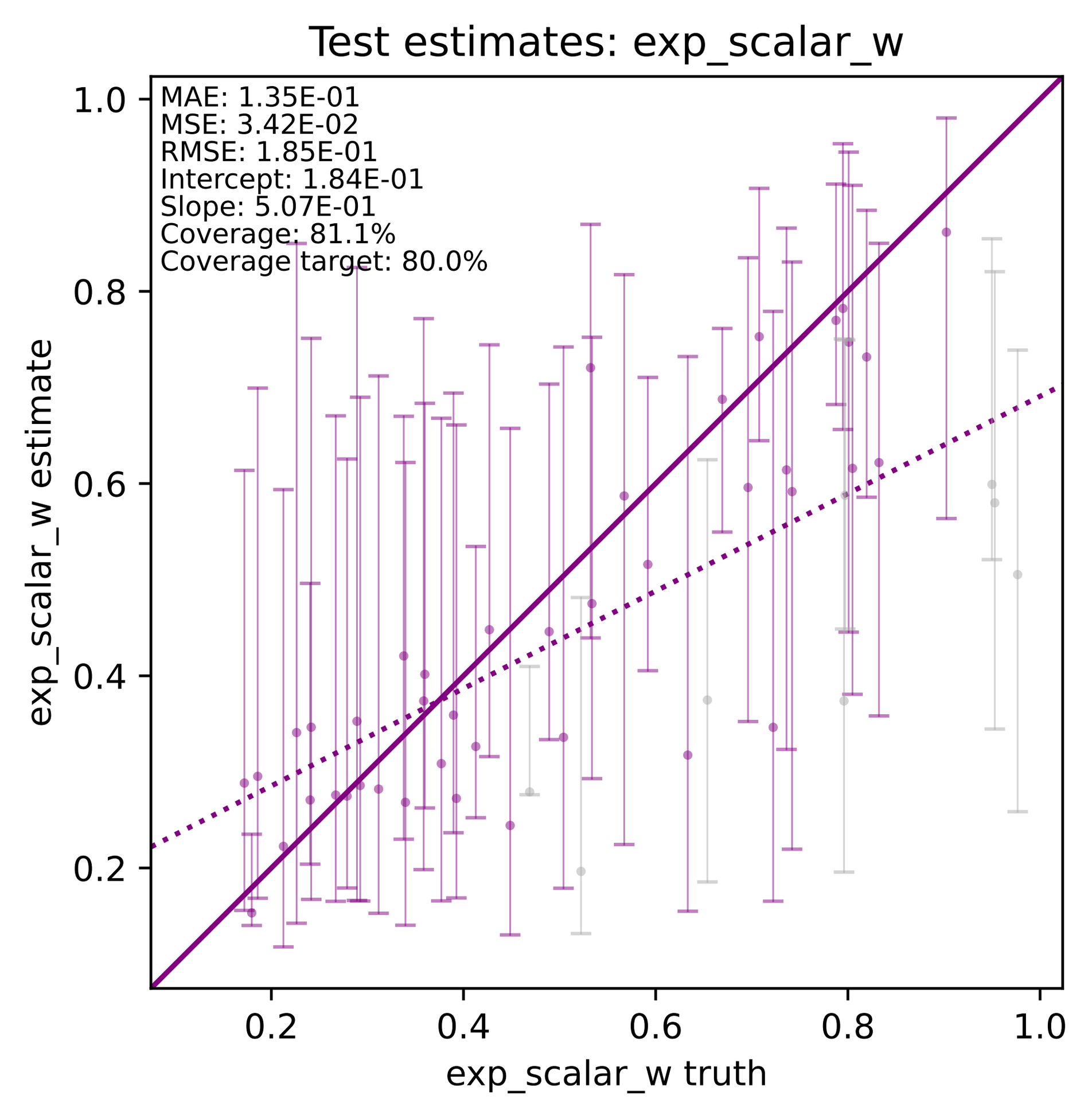

Train estimates: log_base_b
MAE: 6.67E-01
MSE: 1.45E+00
RMSE: 1.21E+00
Intercept: -5.75E-01
Slope: 7.16E-01
Coverage: 79.3%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

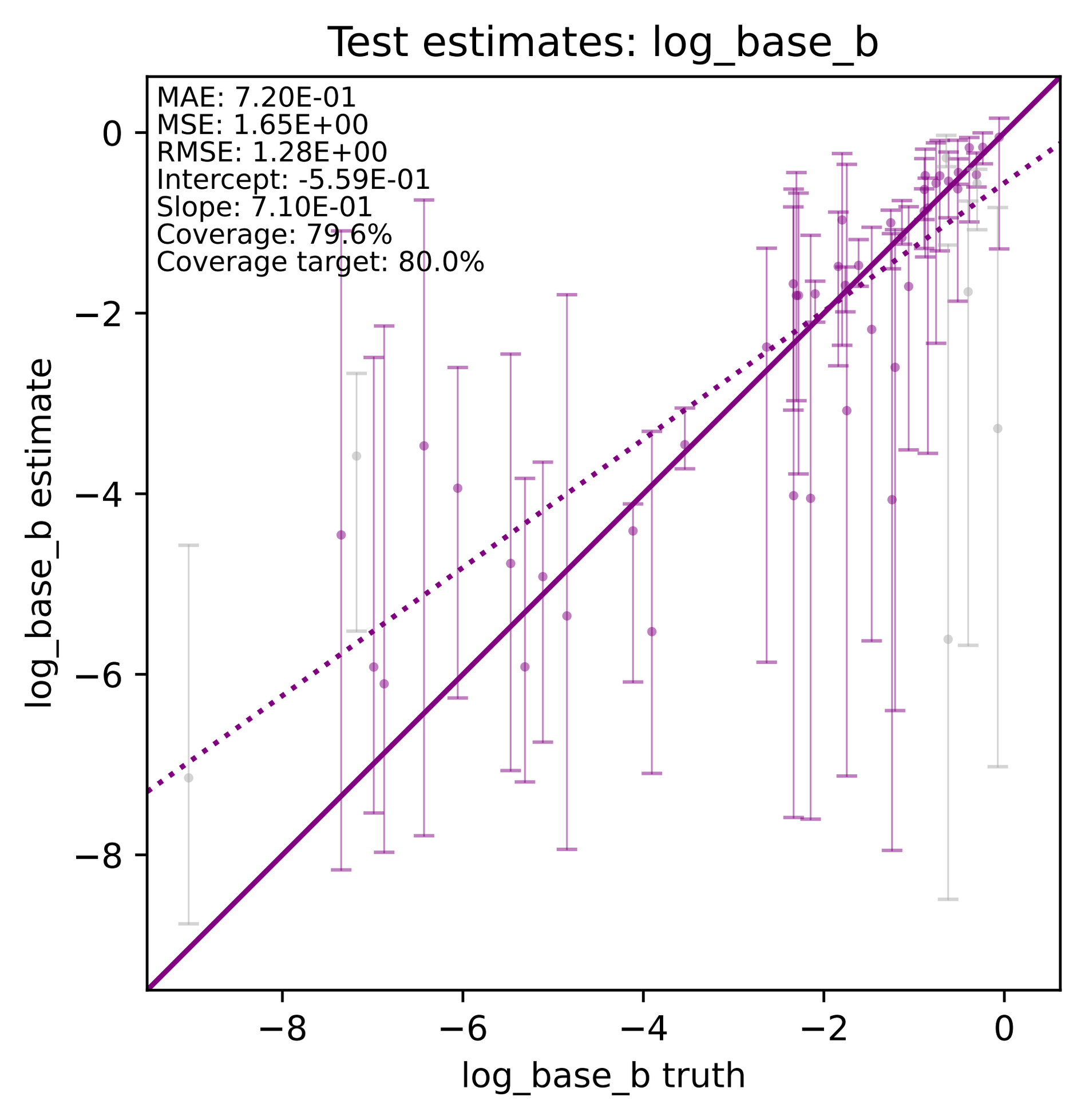

Train estimates: log_base_d
MAE: 2.12E-01
MSE: 1.20E-01
RMSE: 3.46E-01
Intercept: -6.68E-02
Slope: 9.79E-01
Coverage: 79.9%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−1
−2
−3
−4
−5
−6
−7

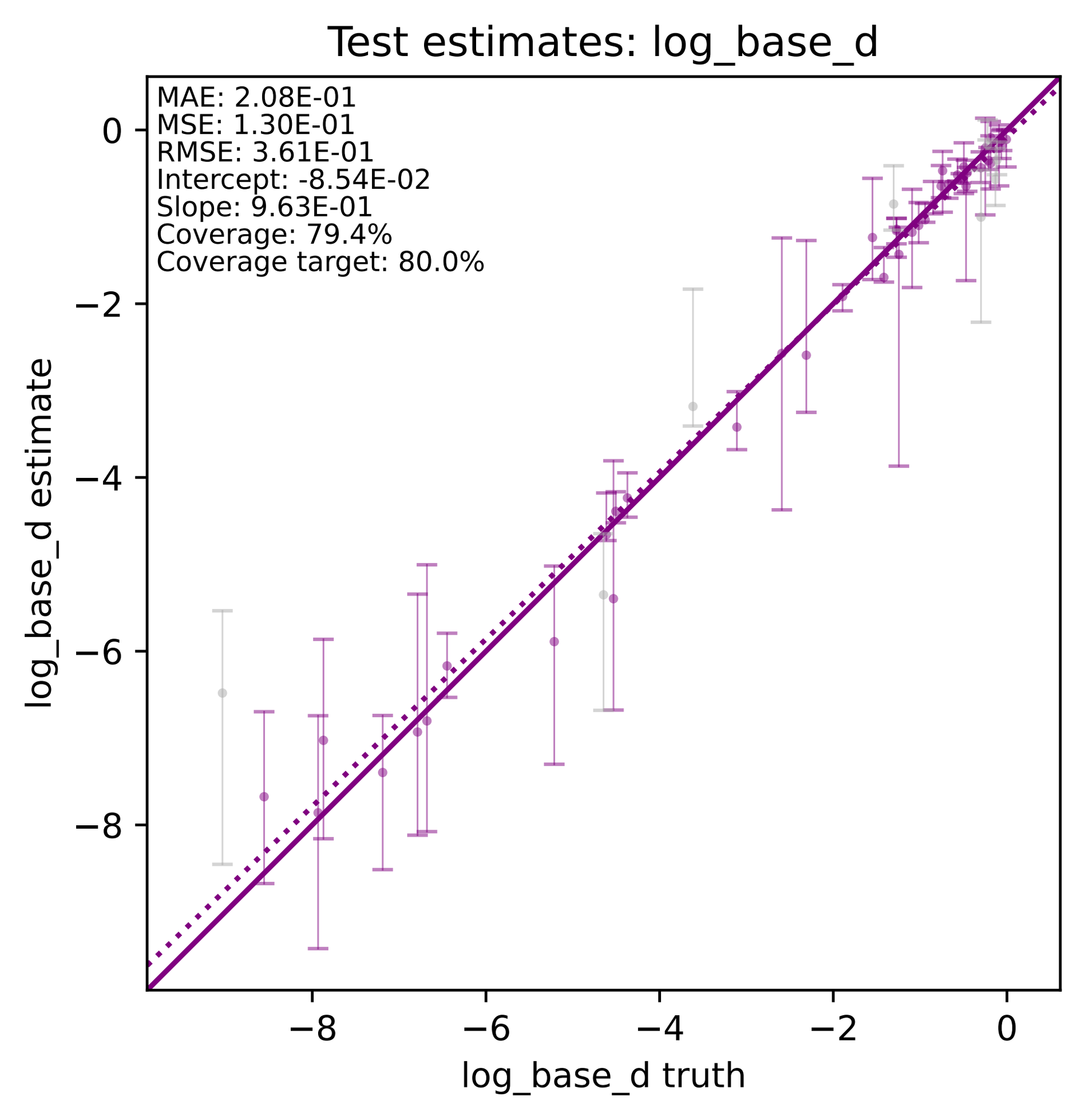

Train estimates: log_base_e
MAE: 7.29E-01
MSE: 8.61E-01
RMSE: 9.28E-01
Intercept: -4.41E-01
Slope: 8.87E-01
Coverage: 80.8%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

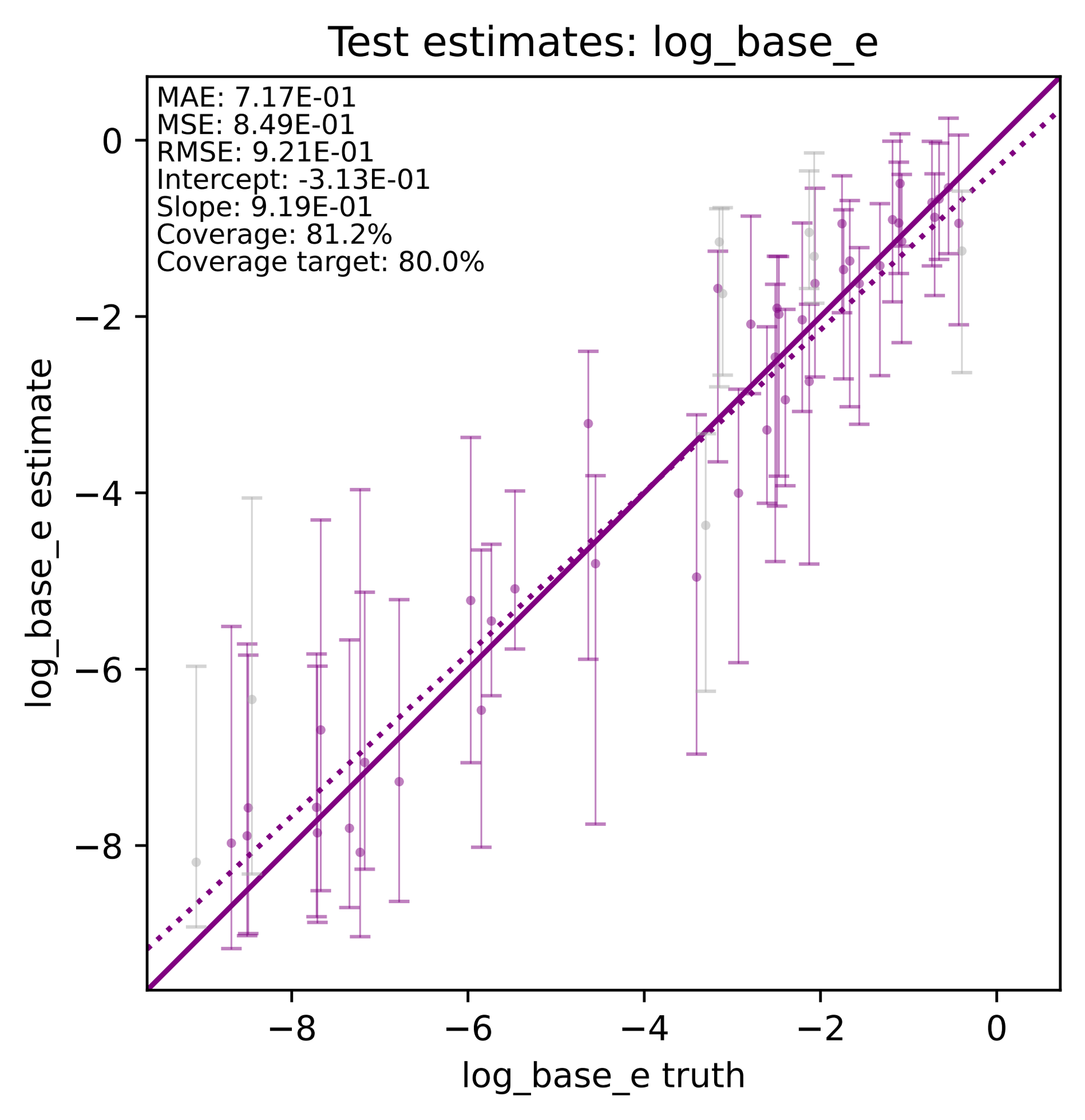

Train estimates: log_base_w
MAE: 7.20E-01
MSE: 1.26E+00
RMSE: 1.12E+00
Intercept: -4.78E-01
Slope: 7.10E-01
Coverage: 80.1%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8

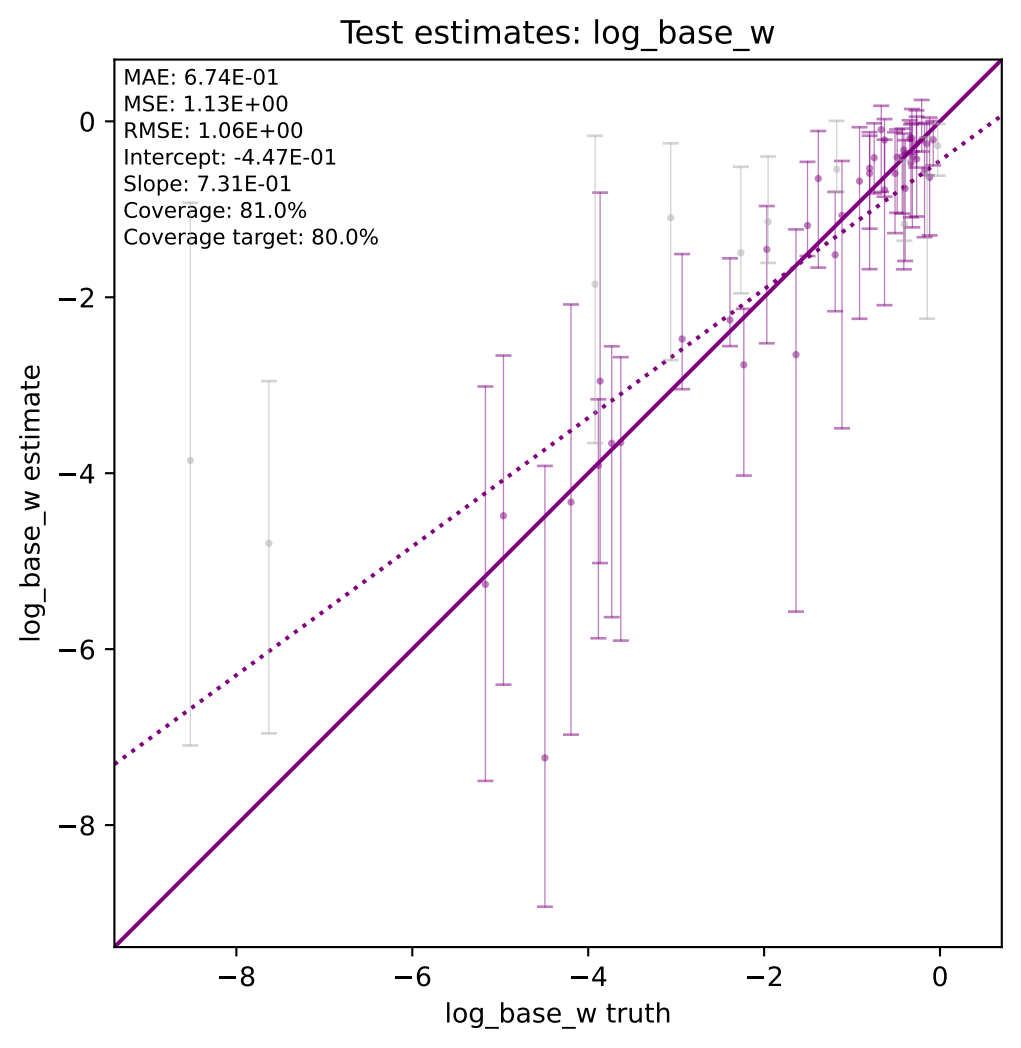

Test estimates: exp_scalar_e
MAE: 1.16E+00
MSE: 2.52E+00
RMSE: 1.59E+00
Intercept: 1.96E+00
Slope: 1.63E-01
Coverage: 81.2%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

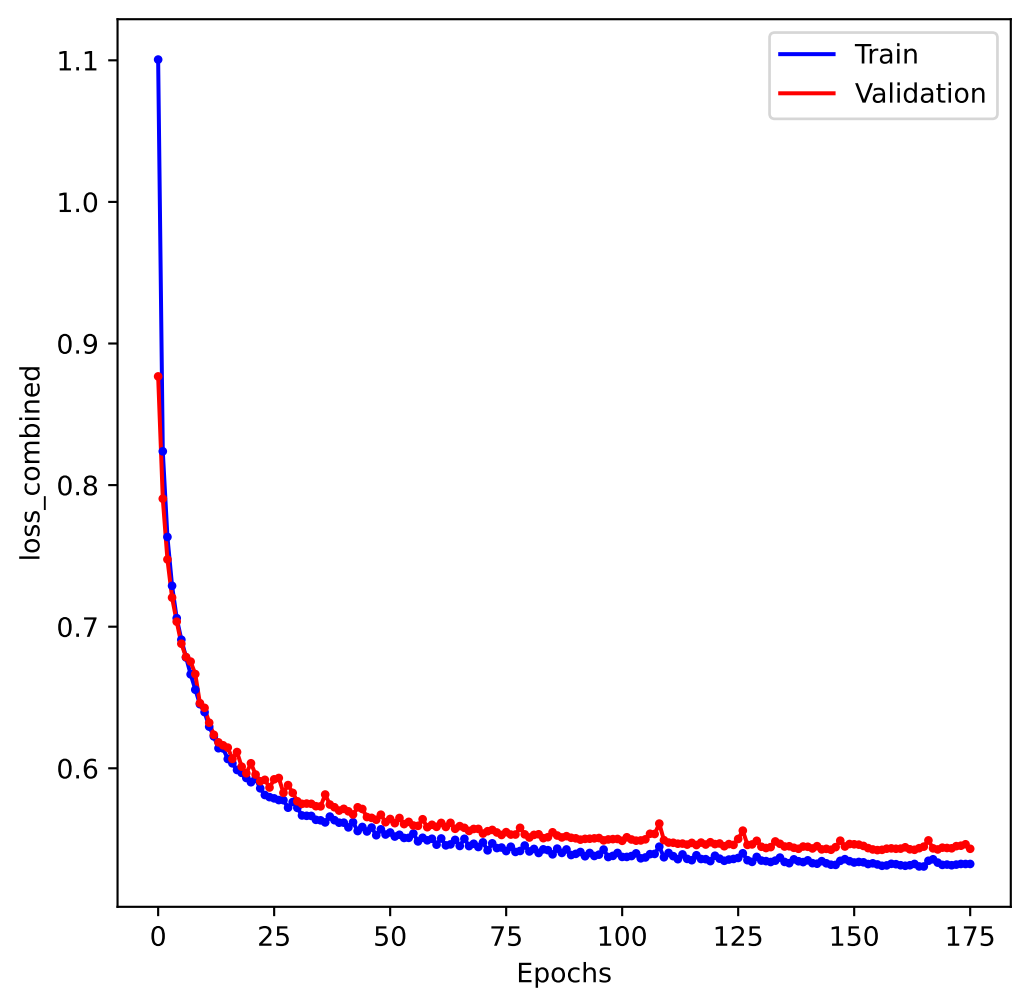

Test estimates: exp_scalar_w
MAE: 1.35E-01
MSE: 3.42E-02
RMSE: 1.85E-01
Intercept: 1.84E-01
Slope: 5.07E-01
Coverage: 81.1%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

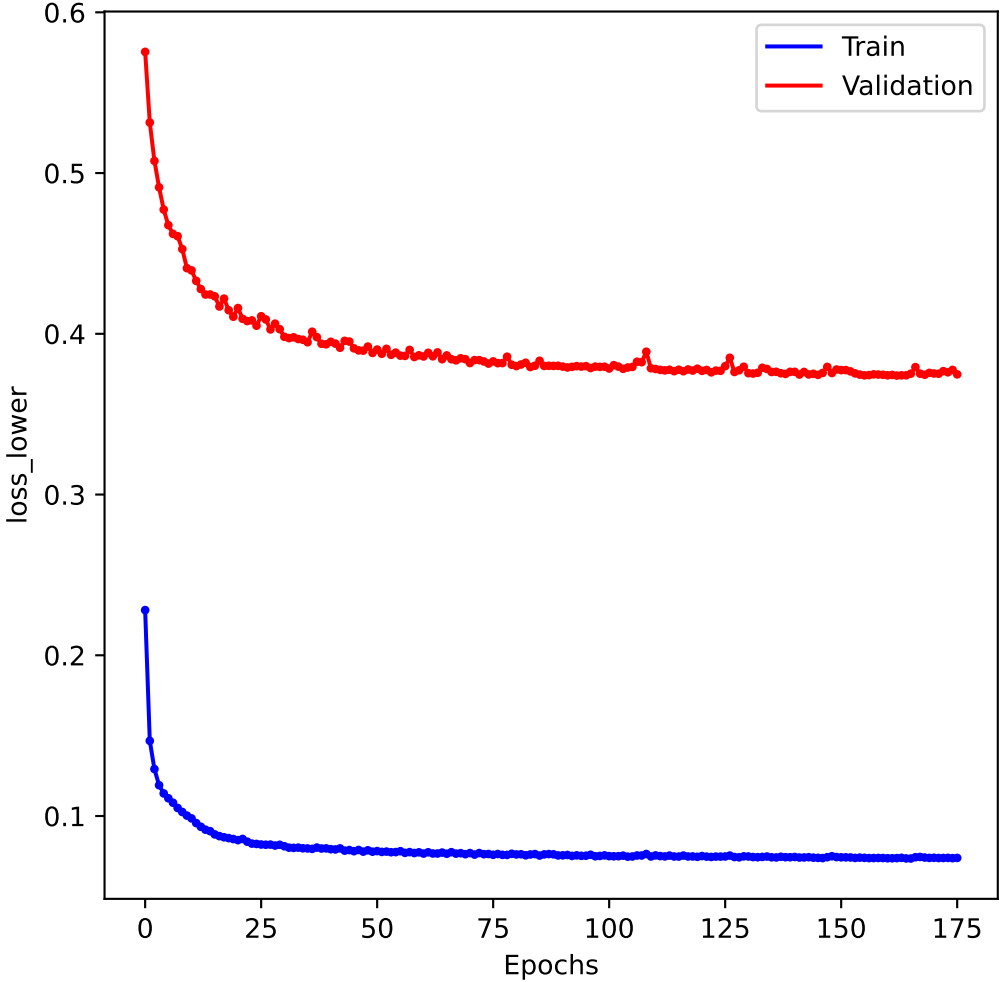

Test estimates: log_base_b
MAE: 7.20E-01
MSE: 1.65E+00
RMSE: 1.28E+00
Intercept: -5.59E-01
Slope: 7.10E-01
Coverage: 79.6%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

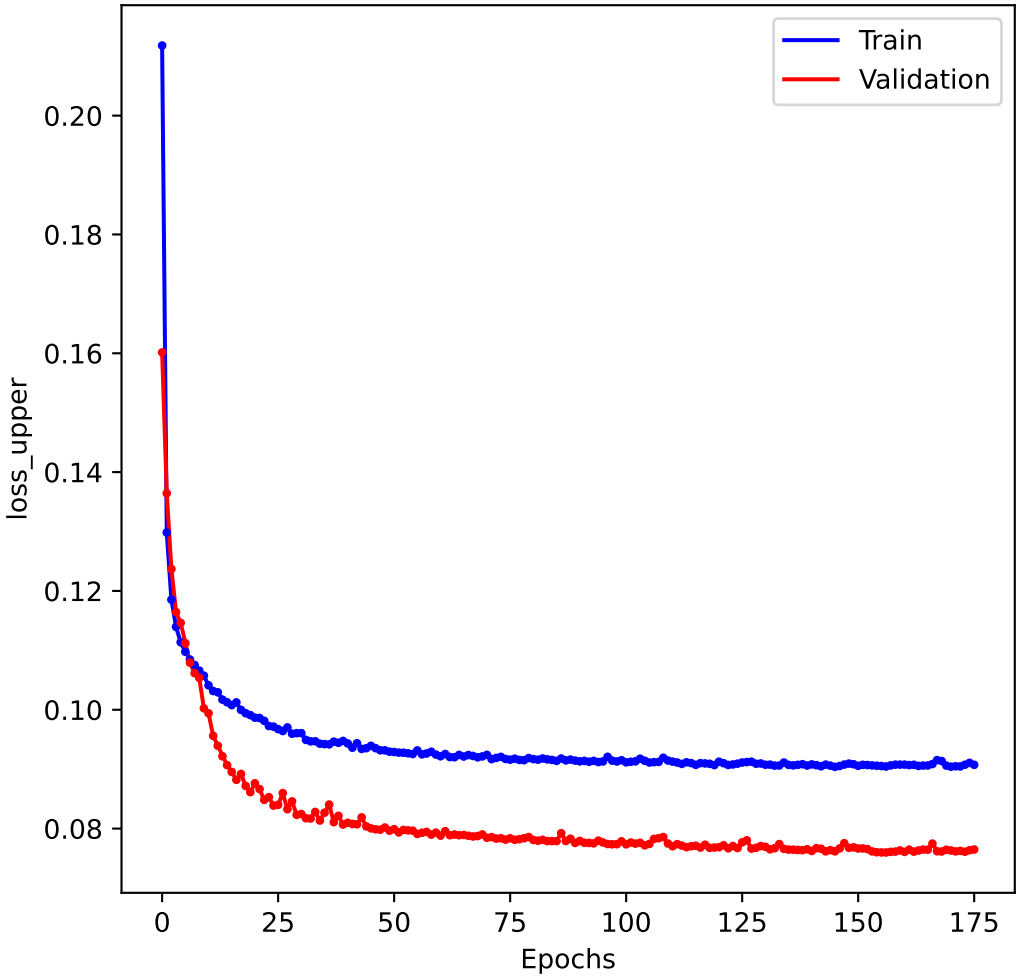

Test estimates: log_base_d
MAE: 2.08E-01
MSE: 1.30E-01
RMSE: 3.61E-01
Intercept: -8.54E-02
Slope: 9.63E-01
Coverage: 79.4%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

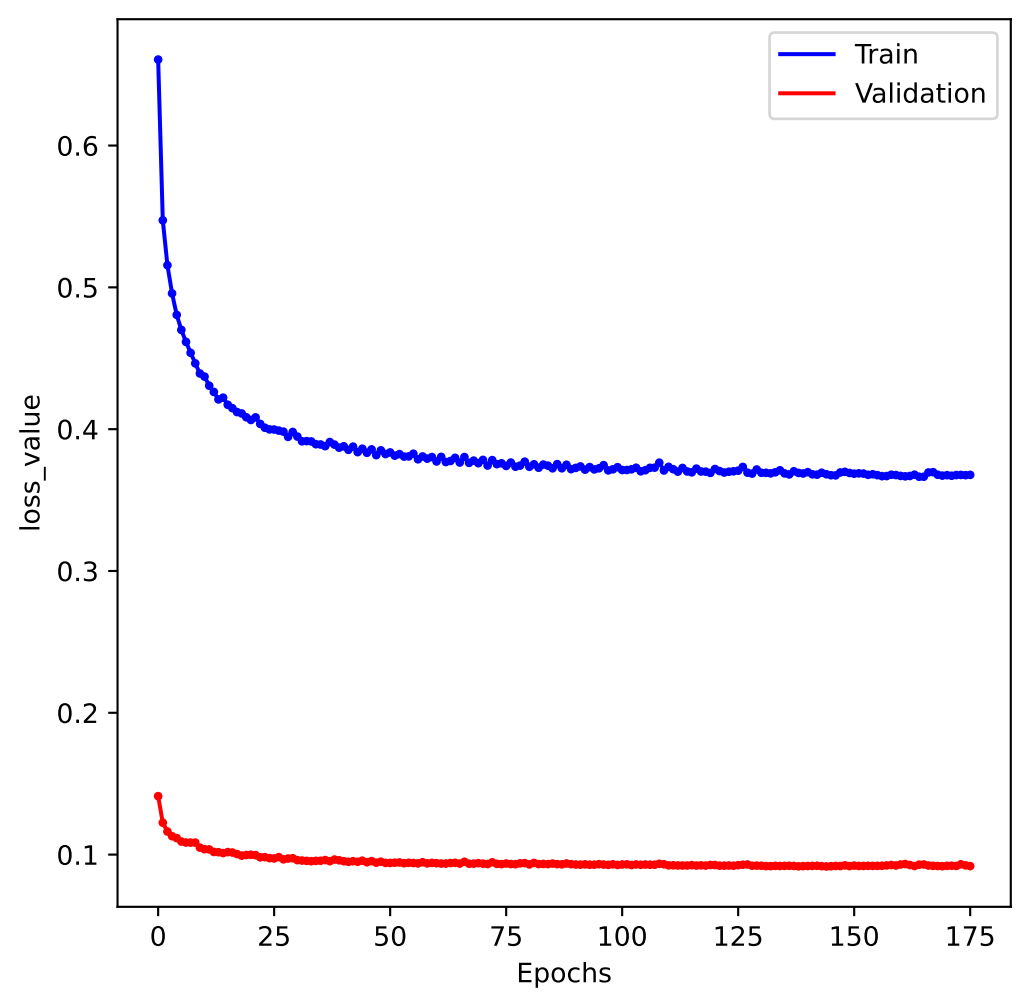

Test estimates: log_base_e
MAE: 7.17E-01
MSE: 8.49E-01
RMSE: 9.21E-01
Intercept: -3.13E-01
Slope: 9.19E-01
Coverage: 81.2%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

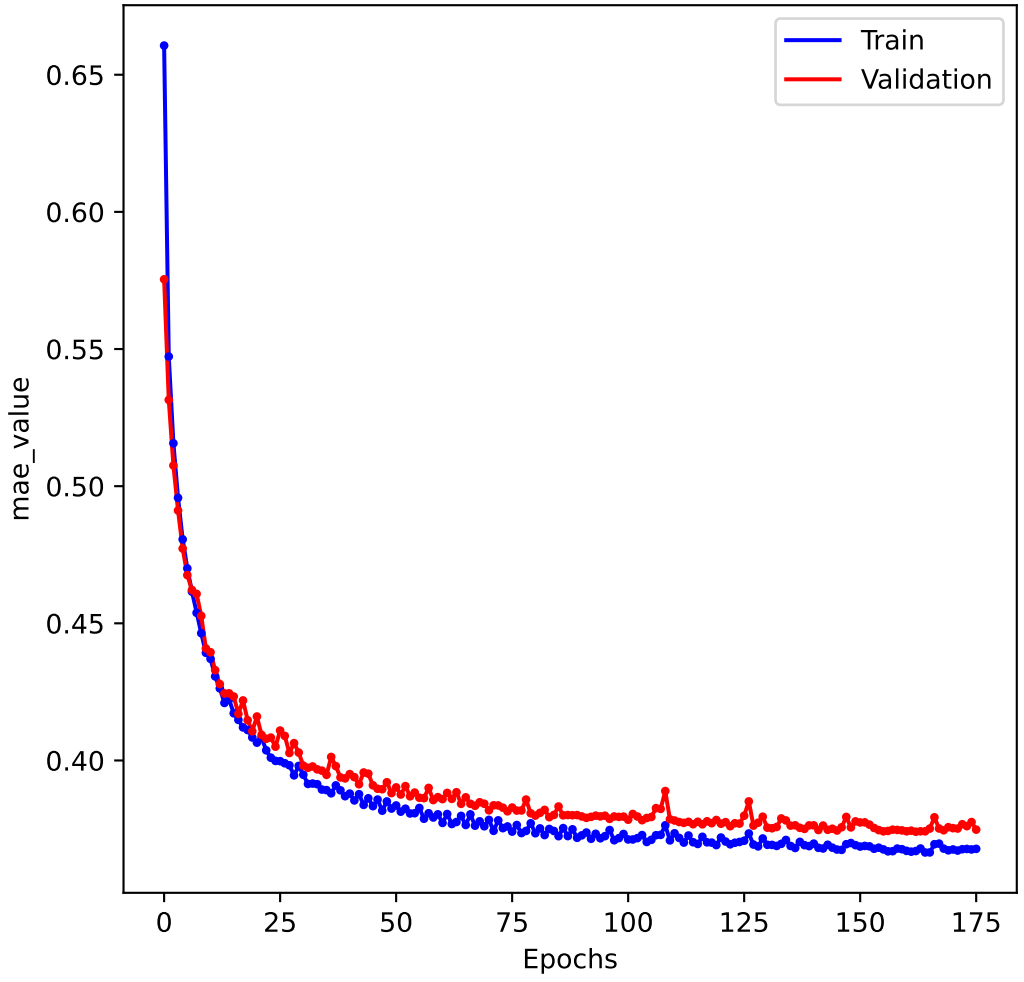

Test estimates: log_base_w
MAE: 6.74E-01
MSE: 1.13E+00
RMSE: 1.06E+00
Intercept: -4.47E-01
Slope: 7.31E-01
Coverage: 81.0%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

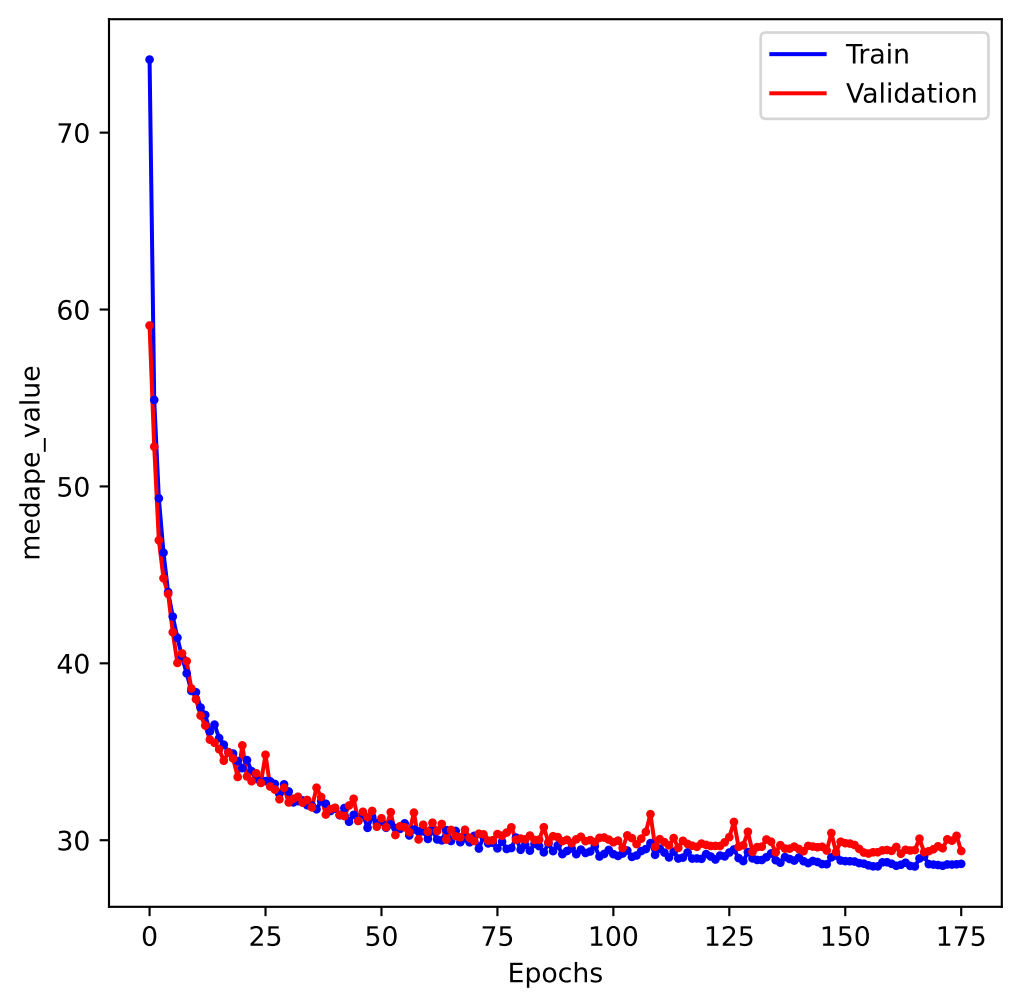

Train
Validation
loss_combined
1.1
1.0
0.9
0.8
0.7
0.6
0
25
50
75
100
125
150
175
Epochs

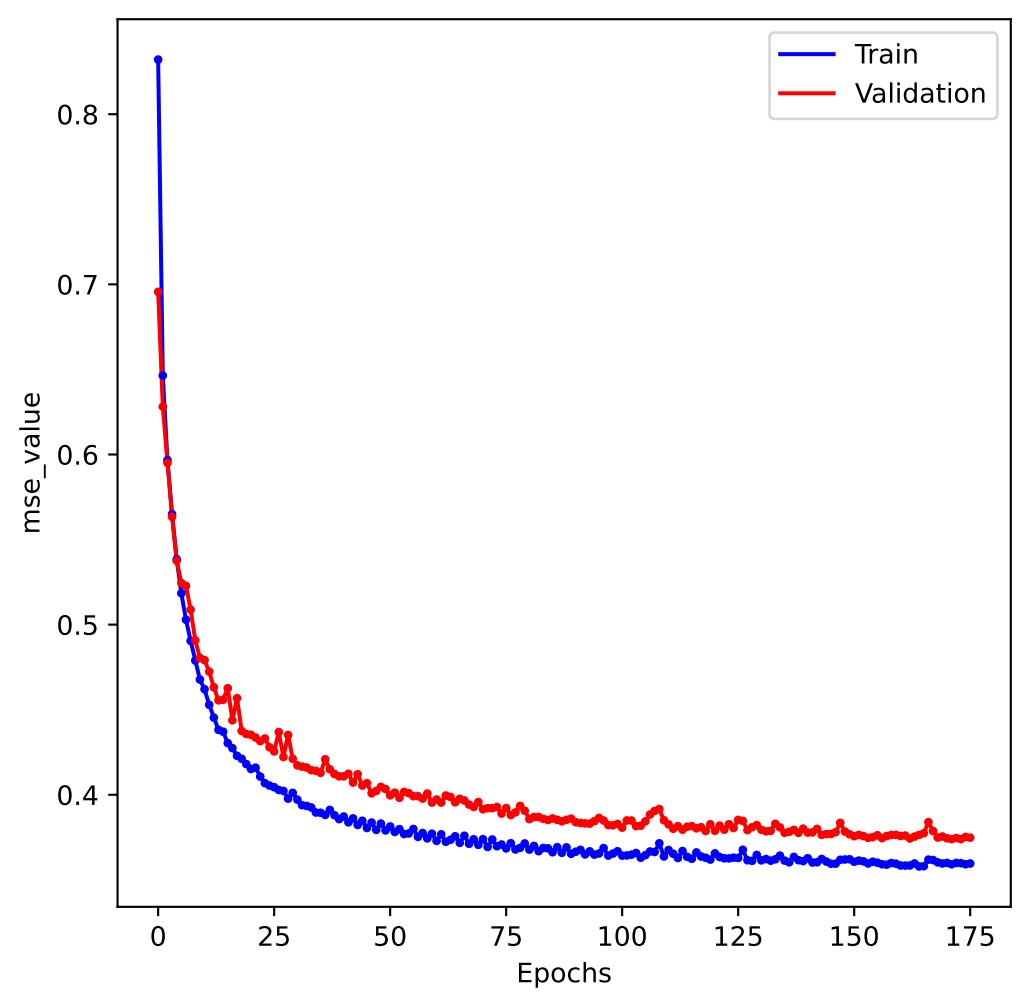

Train
Validation
loss_lower
Epochs
0.6
0.5
0.4
0.3
0.2
0.1
0
25
50
75
100
125
150
175

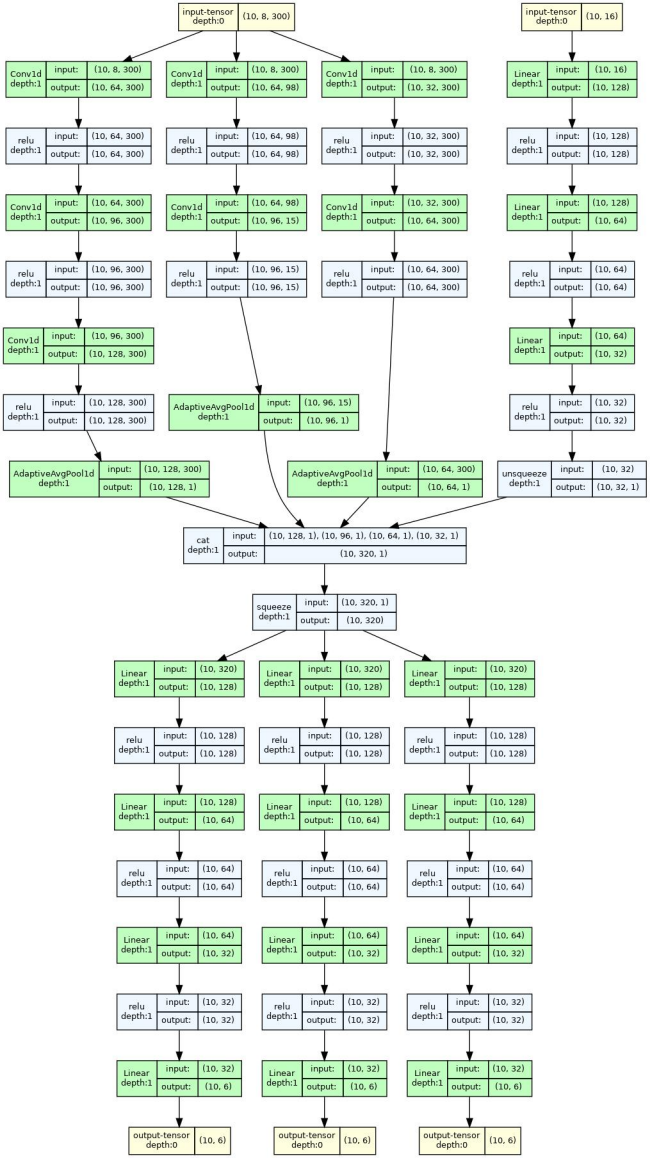

Train
Validation
loss_upper
0.20
0.18
0.16
0.14
0.12
0.10
0.08
0
25
50
75
100
125
150
175
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
25
50
75
100
125
150
175
Epochs


Train
Validation
mae_value
0.65
0.60
0.55
0.50
0.45
0.40
0
25
50
75
100
125
150
175
Epochs


Train
Validation
70
60
50
40
30
medape_value
0
25
50
75
100
125
150
175
Epochs


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0
25
50
75
100
125
150
175
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)